6) RbO_2 is a super oxide which contains Rb+ and O_2^- ions. O_2^- contains one unpaired electron and hence it is paramagnetic.

Option (a)

7) Potassium carbonate cannot be prepared by solvay process. Potassium bicarbonate is fairly soluble in water and does not precipitate out.

Option (c)

8) Li → Mg | Be → Al | B → Si

Option (b)

9) ionic character (difference in electronegativity)

MI < MBr < MCl < MF

Option (b)

10) Castner's process

$NaOH \rightleftharpoons Na^+ + OH^-$

Cathode : $Na^+ + e^- \rightarrow Na$

Anode : $2OH^- \rightarrow H_2O + 1/2O_2 + 2e^-$

Option (a)

11) $CaC_2 + N_2 \xrightarrow{300-350^oC} \underset{\text{Calciumcyanide}}{Ca(CN)_2}$

Option (c)

12) (a) $MgCl_2$

The order of hydration energy of alkaline earth metal is

$Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

13. (p) sodium –yellow (2)

(q) Calcium – Brick red (1)

(r) Barium – apple green (4)

(s) Strontium – Crimson red (5)

(t) Cesium – blue (6)

(u) Potassium – Violet (3)

Option (a)

14. (d)

Among alkali and alkaline earth metals,

K, Rb and Cs alone forms superoxides.

Superoxide O_2^- has 3 electron bond.

15. (a)

both are true and reason is the correct explanation of assertion.

16. Solubility of carbonates decreases down the group and hence the correct order of solubility is,

(b) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

17) **Correct Statement:** Beryllium salts are easily hydrolysed

option (c)

18) Slaked lime $Ca(OH)_2$

The suspension is called milk of lime and the clear solution is called lime water.

Option (c)

19)

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2 \uparrow$$

Na_2CO_3 (Soluble in water) $\xrightarrow{2HCl}$ $2\,NaCl + H_2O + CO_2 \uparrow$

Correct option (b)

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